PROJECT

Voice Agent for Real Estate Inbound and Outbound Leads

System Architecture & Technical Delivery Report

EstateVoice AI | Prepared by Muhammad Hassan - AI Automation Engineer & Full Stack Developer

Project	Voice Agent for Real Estate Inbound and Outbound Leads (product name: EstateVoice AI, example agency: Prime Estates)
Domain	Real estate: property sales and rentals
Technical lead	Muhammad Hassan - AI Automation Engineer & Full Stack Developer
Core platforms	Vapi (voice AI), n8n (workflow automation), GoHighLevel (CRM)
Integration layer	n8n with 7 workflows, connected to Vapi, Twilio, Google Sheets, Google Calendar and GoHighLevel through webhooks and REST APIs
Knowledge layer	Knowledge Base RAG: listings retrieved from a Google Sheet and given to the model as facts before it answers
Scope of delivery	Outbound and inbound voice agents, lead qualification and scoring, calendar booking with SMS confirmation, CRM automation, knowledge base RAG, compliance controls, cost controls, documentation and tests

1. The Project at a Glance

In plain words: This is a phone agent that never sleeps. When someone enquires about a property, it calls them within minutes, asks a few friendly questions, works out how serious they are, books a viewing, and writes everything into the CRM. When people call in, it answers questions about listings from a live knowledge base, and it never makes up facts.

STEP 1	STEP 2	STEP 3	STEP 4	STEP 5	STEP 6	STEP 7
Lead comes in form, ad or list	**Compliance gate** consent, hours, DNC	**AI voice call** Vapi + Twilio	**Qualify and score** needs, hot/warm/cold	**Book viewing** calendar + SMS	**CRM update** fields, stage, notes	**Smart routing** alerts, retries

The seven-step flow every lead follows

7 n8n workflows	**6** live voice tools	**20** CRM custom fields
46 / 46 end-to-end checks passed	**~2.1 s** estimated reply latency	**~$0.11** estimated cost per call minute

Latency and cost are Vapi dashboard estimates for the configured providers.

2. Business Problem and Client Challenges

Real estate agencies lose business in the gap between an enquiry and the first conversation. The project was built around these challenges:

- **Slow speed-to-lead** - leads wait hours for a first call, and by then interest has faded or they have called a competitor.

- **Repetitive qualification work** - agents ask the same six questions on every lead: buy or rent, budget, area, bedrooms, timeline and financing.
- **Inconsistent property answers** - callers get different or wrong details depending on who picks up the phone.
- **Manual CRM data entry** - notes are typed by hand, late or incomplete, so managers cannot trust the pipeline.
- **Missed hot leads** - serious buyers wait in the same queue as casual browsers, because nobody has scored them.
- **Scheduling friction** - booking a viewing takes many messages, and double-bookings happen.
- **Compliance risk and cost** - calling outside allowed hours, after a lead says stop, or calling the same lead twice wastes money and creates risk.

3. Original Plan and What Was Delivered

The project followed the original brief feature for feature. The table lists each planned feature, what it means, and how it was delivered.

Planned feature	What it means	Delivered as
Outbound calling within minutes	A new lead gets a call quickly, with no human step.	Vapi outbound assistant, started by n8n workflow 01 through a webhook.
Lead qualification	Asks buy or rent, budget, area, bedrooms, timeline, financing.	save_qualification tool captures and cleans every answer.
Hot, warm and cold scoring	Ranks leads by how ready they are.	Weighted scoring: hot at 70 or more, warm at 40 or more.
Viewing booking and SMS	Books a real time slot and texts the lead.	check_availability and book_viewing with Google Calendar and Twilio SMS.
CRM automation	Every call is recorded in the CRM.	GoHighLevel fields, stage, tags, notes, summary and transcript.
Hot-lead alerts and transfer	Agents are told at once about serious leads.	Alert SMS plus an optional live-transfer tool.
'Not now' and 'stop calling'	Respects the lead's wishes.	schedule_callback, opt_out, callback scheduler and STOP-reply workflow.
Inbound agent with a knowledge base	Answers property questions from real data.	Knowledge Base RAG: inbound assistant with search_listings over the Listings sheet.
Documented n8n workflows	Anyone can understand and rebuild it.	7 exported workflows, endpoints file, credentials template, validator.
Scripts, prompts and handover	Everything needed to run and maintain it.	System prompts, call script, setup guide, test plan, compliance notes, handover guide.

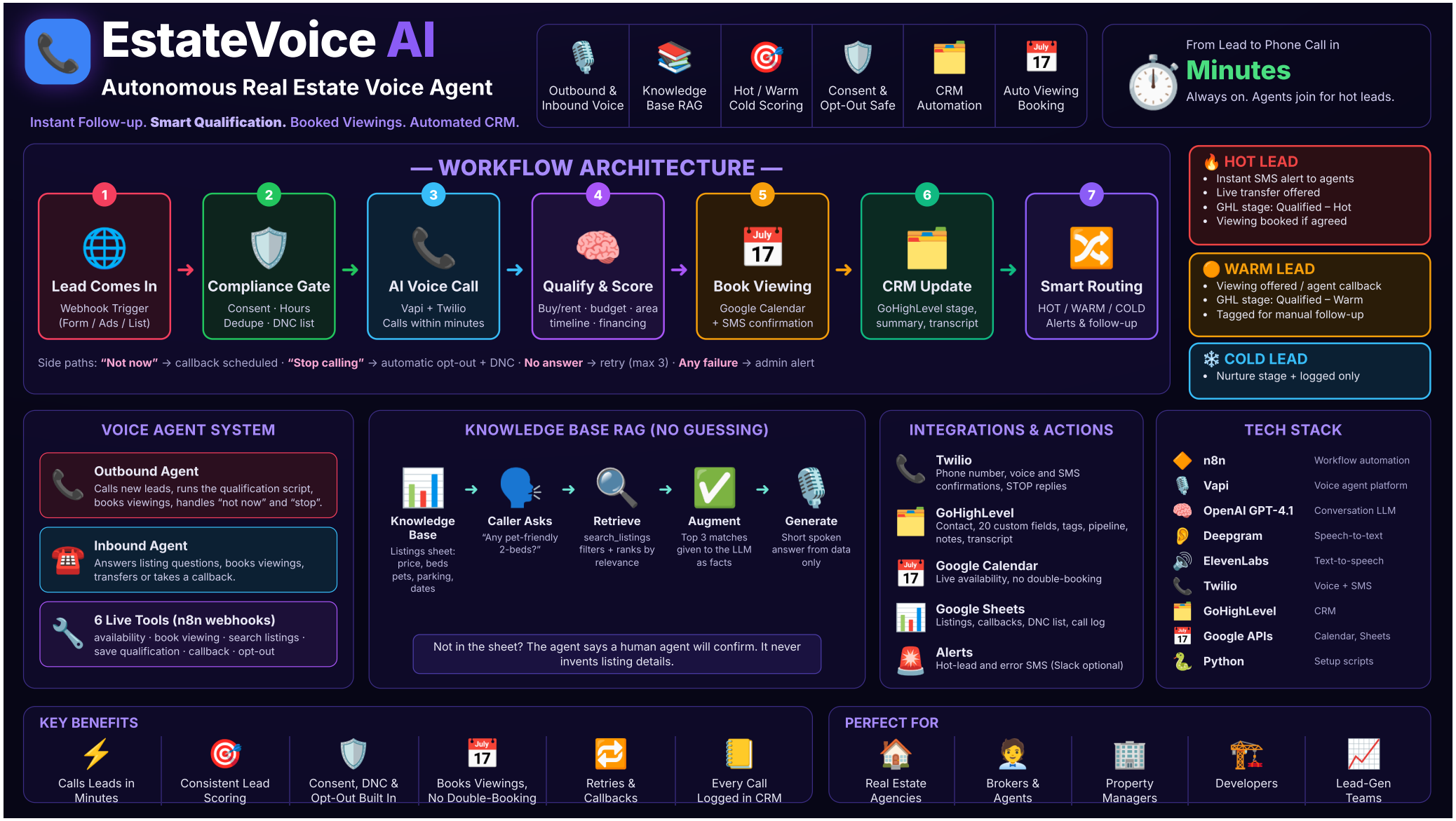
4. Technology Stack: What Each Tool Does

> **In plain words:** Think of the system as a team. Vapi is the voice on the phone, n8n is the brain that decides what happens next, GoHighLevel is the filing cabinet, and Google holds the listings knowledge base and the calendar.

Tool	Role	What we use it for
Vapi	Voice AI platform	Runs the outbound and inbound assistants, tool calling, call recording, voicemail detection and end-of-call summaries.
OpenAI GPT-4.1 mini	Language model	The agent's thinking: understands the caller, turns questions into searches, follows the script and answers from retrieved facts.
Deepgram Nova 3	Speech to text	Turns the caller's speech into text, quickly and accurately.
ElevenLabs	Text to speech	Gives the agent a natural, friendly voice.
Twilio	Telephony and SMS	Phone number for calls, viewing-confirmation texts and STOP replies.
n8n	Workflow automation	Seven workflows: lead intake, tool webhooks (including RAG retrieval), end-of-call report, inbound SMS, callbacks, error handling.
GoHighLevel	CRM	Stores contacts, 20 custom fields, tags, pipeline stages, opportunities, notes and do-not-disturb.
Google Sheets	Knowledge base and data	RAG knowledge base of listings, plus the callbacks queue, do-not-call list and call log.
Google Calendar	Scheduling	Live availability, viewing events and double-booking protection.
Python and Node.js	Tooling	Generate the workflows, provision Vapi, run a mock-service test harness with 46 checks.

5. System Architecture - Product Overview

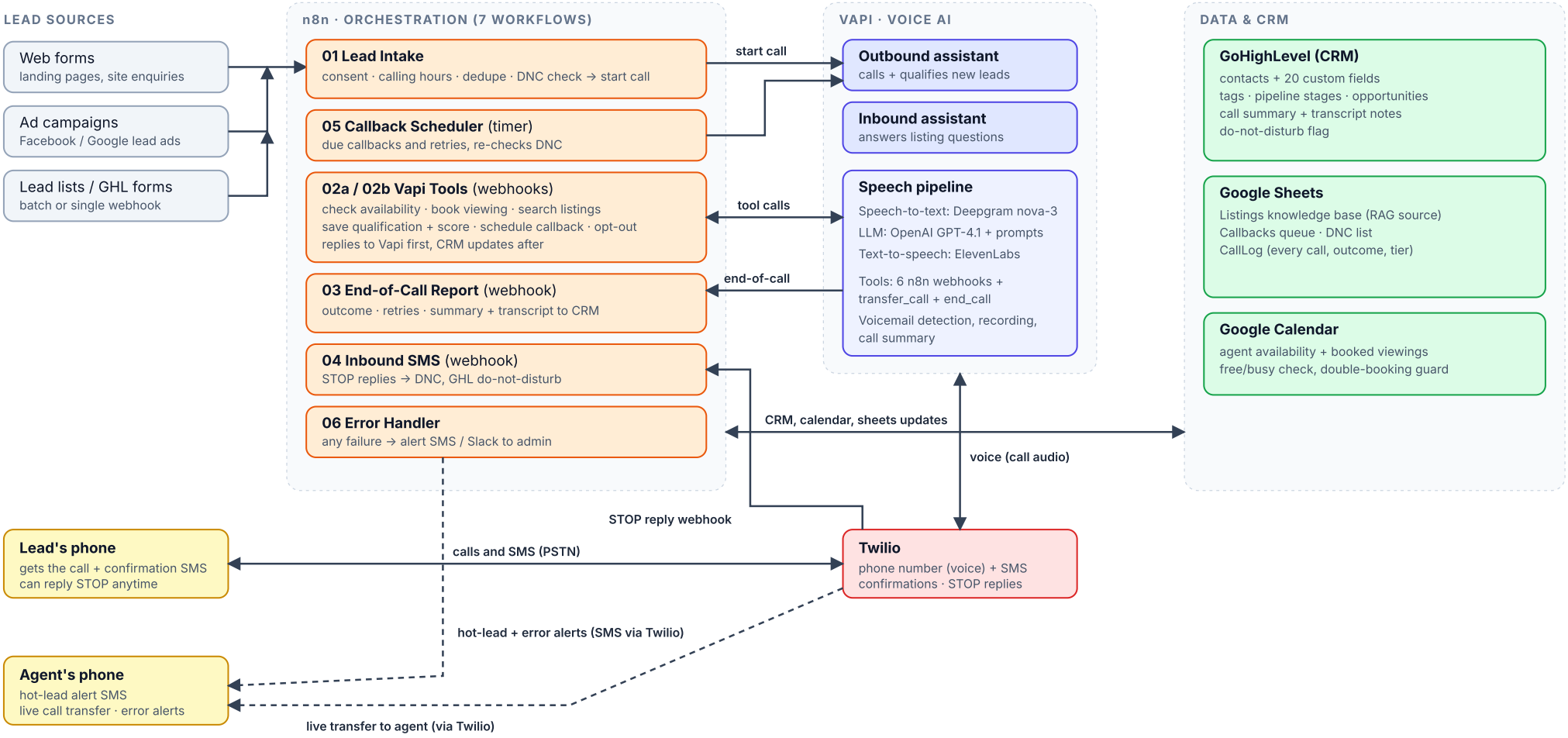
The seven-step workflow, hot, warm and cold routing, the voice agent system, the knowledge base RAG, integrations and the stack.



5. System Architecture - Technical View

Lead sources, the seven n8n workflows, the Vapi voice layer, Twilio, and the data, knowledge base and CRM layer.

Voice Agent for Real Estate Inbound and Outbound Leads — system architecture



STACK

n8n (webhooks, HTTP requests, JavaScript Code nodes, error workflow) · Vapi (voice agent platform) · OpenAI GPT-4.1 · Deepgram · ElevenLabs
Twilio (voice + SMS) · GoHighLevel API v2 (CRM) · Google Calendar and Sheets APIs · Python scripts for setup and workflow generation

6. The Voice Agent System

> **In plain words:** There are two assistants. The outbound one calls new leads and qualifies them. The inbound one answers callers who ask about properties. Both use the same six tools to look things up and take action.

6.1 Two assistants

- **Outbound assistant (Ava)** - calls new leads, runs the qualification script, books viewings, and handles 'not now' and 'stop calling'. Detects voicemail and leaves a short message.
- **Inbound assistant (Ava)** - answers listing questions through the knowledge base RAG, books viewings, takes callbacks, and hands over to an agent during office hours.

6.2 The six live tools

Tool	What the agent does with it	Handled by
check_availability	Looks up open viewing slots. Never invents times.	Workflow 02a
book_viewing	Creates the calendar event and sends the SMS confirmation.	Workflow 02a
search_listings	The RAG retriever: finds active listings by area, price, bedrooms and type.	Workflow 02a
save_qualification	Saves what was learned, scores the lead and writes it to the CRM.	Workflow 02b
schedule_callback	Schedules a follow-up call at the time the lead asks for.	Workflow 02b
opt_out	Adds the number to the do-not-call list and marks the CRM contact.	Workflow 02b

6.3 What makes the agent dependable

- **Grounded answers.** Property facts come only from the knowledge base. If something is missing, the agent says an agent will confirm.
- **Natural voice style.** Short sentences, one idea at a time, at most two properties per reply, prices said the way people say them.
- **Honest about being AI.** It introduces itself as a virtual assistant and mentions recording at the start of the call.
- **Prepared for real calls.** Rules cover callbacks, opt-out, requests for a human, other cities, unclear speech, silence, rude callers and sold properties.
- **Consistent.** Lower model temperature, a silence timeout and a maximum call length keep every call predictable.

7. Knowledge Base RAG

> **In plain words:** Before the agent answers a property question, it looks the answer up. It searches the listings knowledge base, takes the best three matches, and speaks only from those facts. This is retrieval-augmented generation (RAG), and it is why the agent does not invent prices or features.

7.1 How a question is answered

Stage	What happens	Component
1. Ask	The caller asks, for example: "Any pet-friendly two-bed rentals in Riverside under twenty-five hundred?"	Deepgram
2. Understand	The model turns the question into a structured search: rent, at least 2 bedrooms, up to $2,500, Riverside, pets.	GPT-4.1 mini
3. Retrieve	search_listings reads the Listings sheet and keeps active listings that match type, bedrooms, price range, area and property type.	n8n + Google Sheets
4. Rank	Results are ranked by keyword relevance to the question, then by price.	n8n

Stage	What happens	Component
5. Augment	Only the top 3 matches go back to the model as compact facts: price, bedrooms, bathrooms, size, pets, parking, date, highlights.	Vapi tool result
6. Generate	The model gives a short spoken answer using only those facts.	GPT-4.1 mini + ElevenLabs

7.2 Knowledge base content

- **One row per property** in a Google Sheet: id, status, sale or rent, title, address, neighborhood, city, price, bedrooms, bathrooms, size, property type, pets, parking, available from and highlights.
- **Live updates.** The sheet is read on every search, so a price change or a new listing is used on the very next call. No re-indexing.
- **Status aware.** Only active listings are offered. Sold or on-hold properties are never presented as available.

7.3 Guardrails

- If nothing matches, the agent says so and offers the closest option, naming what differs.
- If a named property is not found, the search returns nothing, never a "similar" listing passed off as the same one.
- "No listings" is only said when the search really returned zero matches.
- Facts that are not in the knowledge base (deposit, fees, schools) are handed to a human agent.

7.4 Why structured retrieval

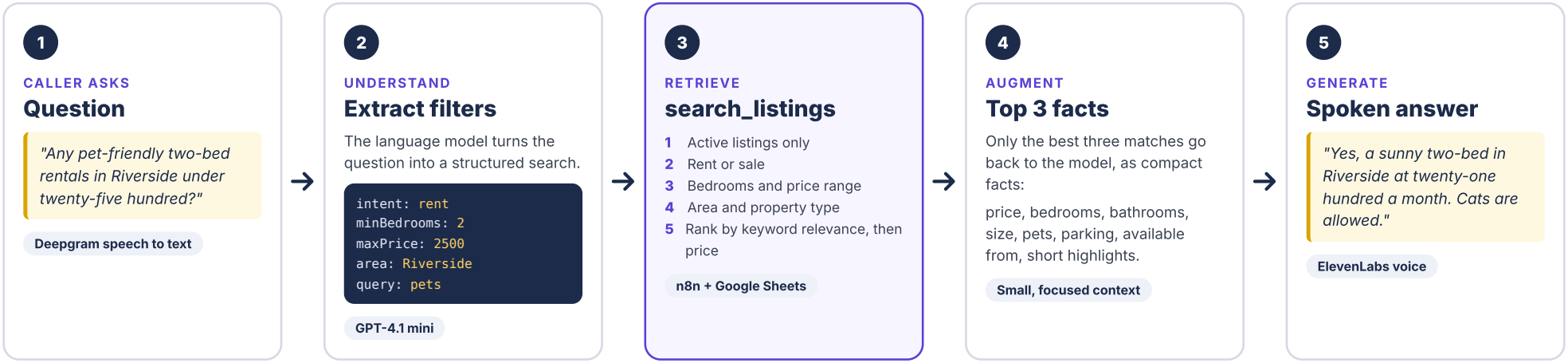
Listings are structured data with exact numbers, so the retriever filters on real fields and ranks by relevance instead of using a vector database. Price and bedroom filters are exact, there are no embedding or database fees, and the knowledge base stays a simple sheet that the agency's team can edit.

7. Knowledge Base RAG - Diagram

From the caller's question to a grounded spoken answer, with the guardrails that prevent guessing.

Knowledge Base RAG: how the agent answers property questions

Retrieval-augmented generation. The agent retrieves facts from the listings knowledge base first, then speaks only from those facts.



Guardrails: the agent never guesses

- **No match:** says so honestly and offers the closest option.
- **Named property not found:** returns nothing, never a "similar" listing.
- **Fact not in the data** (deposit, fees, schools): "an agent will confirm".
- **"No listings"** is only said when the search really returned zero.

Why structured retrieval for listings

- **Exact numbers:** price and bedroom filters are precise, not approximate.
- **No vector database or embedding costs** to run or maintain.
- **Always current:** the team edits a Google Sheet; the next call reads it.
- **Fast and cheap:** three short records keep the prompt small.

8. n8n Automation Engine: Seven Workflows

n8n connects the voice agent to everything else. Each workflow has one clear job.

#	Workflow	Starts when	What it does
01	Lead intake	A form, ad or list sends a lead	Cleans the phone number, checks consent, calling hours, duplicates and do-not-call, then starts the Vapi call.
02a	Vapi tools: calendar and listings	The agent calls a tool during a call	Check availability, book viewing and RAG listing search. Replies to Vapi first, then does slower work.
02b	Vapi tools: leads and compliance	The agent calls a tool during a call	Save qualification and score, schedule callback, opt out.
03	End-of-call report	A call ends	Records outcome, summary and transcript, updates the CRM, queues retries and alerts agents.
04	Inbound SMS	A lead texts STOP	Adds the number to do-not-call and sets do-not-disturb in the CRM.
05	Callback scheduler	A timer, every 5 minutes	Starts due callbacks and retries, re-checking do-not-call first.
06	Error handler	Any other workflow fails	Alerts the administrator by SMS or Slack.

9. GoHighLevel CRM Integration

> **In plain words:** The CRM is where the sales team works. After every call, the contact is already filled in, placed in the right stage, and has a written summary, so agents start from facts and not from scratch.

9.1 Seven-stage pipeline: AI Lead Follow-up

Stage	Operational purpose
New Lead	Lead arrived and passed the compliance gate.
Contacted - No Answer	A call was placed with no answer. A retry is queued.
Qualified - Hot	Score of 70 or more. An agent is alerted immediately.
Qualified - Warm	Score of 40 or more. Tagged for manual follow-up.
Nurture - Cold	Lower score. Logged for nurture.
Viewing Booked	A viewing was confirmed in the calendar.
Do Not Contact	The lead asked to stop, or replied STOP by SMS.

9.2 Twenty custom fields written for every contact

Group	Fields
Lead needs	AI Intent, Budget Min, Budget Max, Preferred Area, Bedrooms, Timeline, Financing
Scoring	AI Lead Score, AI Lead Tier, AI Lead Source
Call record	AI Call Summary, AI Notes, Last Call Id, Last Call At, Call Attempts
Follow-up	AI Next Callback At, AI Opt Out
Viewing	AI Viewing Datetime, Viewing Event Id, Viewing Call Id

- **Also written:** tier and outcome tags, a note with the summary and full transcript, an opportunity in the right stage, and the do-not-disturb flag on opt-out.
- **No duplicates:** contacts are matched by phone number, so a returning lead updates the same record.

10. Lead Scoring and Smart Routing

In plain words: Every answer earns points. More points mean a more ready buyer. The score decides who gets a human call first.

Signal	Points awarded
Timeline (up to 35)	Immediately 35, within 1 month 30, 1 to 3 months 20, 3 to 6 months 8, 6 months or more 3, just browsing 0
Financing (up to 25)	Cash 25, pre-approved 25, ready to apply 22, mortgage in progress 14, not started 5, not ready 0
Specificity (up to 20)	Gave a budget 8, an area 7, bedrooms 5
Engagement (up to 35)	Interested in a viewing 15 (maybe 6), viewing booked 15, wants an agent 5

Tier	Score	What happens automatically
Hot	70 or more	Instant SMS alert to agents, stage Qualified - Hot, live transfer offered, viewing booked if agreed.
Warm	40 to 69	Viewing or callback offered, stage Qualified - Warm, tagged for manual follow-up.
Cold	Below 40	Nurture stage and logged only.

11. Cost Efficiency

In plain words: A typical 3-minute qualification call costs about 33 cents. The system is designed so that money is only spent on real conversations with leads who agreed to be called, and every call is kept short and focused.

~$0.11	~$0.33	~$33
per call minute	per 3-minute call	per 100 three-minute calls

Estimates based on the Vapi dashboard figures for the configured providers. Actual cost depends on call length and provider pricing.

11.1 Cost per minute, by component

Component	Provider	Estimated cost per minute
Call orchestration	Vapi platform	about $0.05
Voice	ElevenLabs	about $0.036
Speech to text	Deepgram Nova 3	about $0.01
Language model	OpenAI GPT-4.1 mini	about $0.01
Total		about $0.11

11.2 How the design keeps cost low

Cost lever	How it saves money
Right-sized model	GPT-4.1 mini is fast and low-cost, and ample for a scripted qualification call.
Small RAG context	Only the top 3 listings are sent to the model, so prompts stay short and replies fast.
No vector database	The knowledge base is a Google Sheet, so there are no embedding or database fees.
No wasted calls	Consent, calling hours, a 10-minute duplicate check and do-not-call run before any call is placed.
Voicemail detection	The agent leaves a short message and hangs up instead of talking to a machine.
Capped retries	At most 3 attempts, spaced 2, 24 and 48 hours apart, so no lead is called endlessly.
Call limits	A 4-minute maximum per call and a silence timeout stop runaway calls.
Short voice replies	Brief sentences mean fewer seconds of speech to pay for.
Reply-first tools	Tool webhooks answer Vapi immediately and do slower CRM work afterwards, so calls do not stall.
Free and trial tiers	Built on n8n Cloud, Google Sheets and Google Calendar, with no paid database or servers to run.

11. Cost Efficiency - Diagram

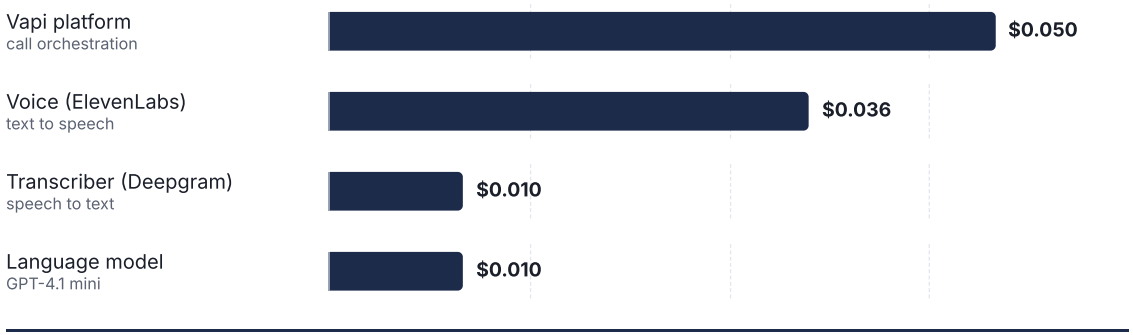
Estimated cost per call minute by component, cost per call, and the design choices that keep it low.

Cost efficiency: about $0.11 per call minute

What one minute of an AI call costs, by component, and the design choices that keep it low.

Estimated cost per call minute, by component

US dollars per minute



Total per minute — **about $0.11**

$0.33
per 3-minute call
A typical qualification call: needs, score, viewing offer.

$33
per 100 calls
100 leads called and qualified at 3 minutes each.

$0
extra for the knowledge base
Listings live in a Google Sheet. No vector database or embedding fees.

4 min
hard cap per call
Plus a silence timeout, so no call can run up a bill.

Where the savings come from

Right-sized model
GPT-4.1 mini: fast and low-cost, ample for a scripted call.

Small RAG context
Only the top 3 listings go to the model, so prompts stay short.

No wasted calls
Consent, calling hours, 10-minute dedupe and do-not-call checks run first.

Voicemail detection
A short message and hang-up, not minutes of talking to a machine.

Capped retries
At most 3 attempts, spaced 2, 24 and 48 hours apart.

Short voice replies
Brief sentences mean fewer seconds of speech to pay for.

Efficient workflows
Tools reply to Vapi first; slow CRM work runs after, so calls stay short.

Free and trial tiers
n8n Cloud, Google Sheets and Calendar, with no paid database.

Per-minute figures are estimates from the Vapi dashboard for the configured providers. Actual cost depends on call length and provider pricing.

12. Compliance and Reliability Controls

Control	Setting
Consent	A consent flag is required before any outbound call.
Calling window	09:00 to 20:00 in the lead's own timezone, Monday to Saturday.
Duplicate protection	The same lead within 10 minutes is ignored.
Retries	Up to 3 attempts, after 2 hours, 24 hours and 48 hours.
Opt-out	Immediate. Do-not-call list, CRM flag, and pending callbacks cancelled.
Scheduler	Checks every 5 minutes, up to 5 calls per run, 8 seconds apart.
Viewing slots	10:00 to 18:00, 45-minute slots, 15-minute buffer, 3 hours notice, 14 days ahead.
Failure handling	The error workflow alerts the administrator. Tool replies never block the call.

13. Delivery and Verification

What was handed over

- **Voice agents:** outbound and inbound Vapi assistants with six tools plus transfer and end-call, provisioned by script.
- **Automation:** 7 n8n workflow files, an endpoints file, a credentials template and a workflow validator.
- **Knowledge base:** a Google Sheet with Listings (the RAG source), Callbacks, DNC and CallLog tabs.
- **Prompts and scripts:** outbound and inbound system prompts, a call script, and handling rules for real-life situations.
- **Documentation:** setup guide, reference, test plan, compliance notes and handover guide.

How it was verified

- **46 of 46 end-to-end checks passed.** The workflows ran on a real n8n instance against local stand-ins for GoHighLevel, Vapi, Twilio, Google Calendar and Google Sheets.
- **RAG retrieval tests** check that filters apply and that an unknown property reference returns no matches instead of a wrong listing.
- **Workflow validator** confirms all seven exported workflows are structurally correct.
- **Live voice test calls** in Vapi covered listing questions, follow-ups on pets and parking, and questions the data cannot answer.

Engineering approach

- **Workflows as code.** One Python toolkit generates all workflows from a single configuration file, so the whole system can be rebuilt in seconds.
- **One-command provisioning.** A script creates or updates the Vapi tools and both assistants through the API.
- **Built on free and trial plans** of every tool in the stack.

14. Key Capabilities and Operational Impact

Metric	What the system delivers
Speed to first touch	Automatic call within minutes of the lead arriving, with no human step.
Reply speed on the call	About 2.1 seconds estimated from the end of speech to the agent's reply.
Answer accuracy	Property answers come from the knowledge base RAG, never from guesswork.
Cost of a qualification call	About $0.11 per minute, so roughly $0.33 for a 3-minute call.
Lead prioritisation	Every lead scored and tiered hot, warm or cold the moment the call ends.
CRM data entry	20 fields, stage, summary and transcript written automatically, with no manual typing.
Viewing booking	Booked on the call from live availability, with double-booking protection and an SMS.
Follow-up	No-answers retried automatically up to 3 times, and callbacks kept without reminders.
Compliance	Consent, calling hours, do-not-call and opt-out enforced on every lead and call.
Quality	46 of 46 automated checks passed.

Prepared by Muhammad Hassan - AI Automation Engineer & Full Stack Developer. This document summarises the architecture and technical delivery of the Voice Agent for Real Estate Inbound and Outbound Leads project. Latency and cost figures are platform estimates, not guarantees.